

Google Looker Studio Learning Documentation

Google Looker Studio (Data Studio)

Introduction to Google Looker Studio

Looker Studio is a cloud-based business intelligence and data visualisation platform used to convert raw data into meaningful reports and interactive dashboards.

It allows users to connect multiple

- data sources
- analyse information
- create visual representations and
- communicate insights effectively with stakeholders.

In the data analysis workflow, Looker Studio works as a reporting and visualisation layer where cleaned and structured data can be transformed into dashboards that support business decision making.

Working of Looker Studio

Overall workflow of looker studio follows the process of:

Data Source > Connector > Data Preparation > Analysis > Visualization > Reporting > Sharing

The process starts by connecting data from different sources using connectors. After connecting the data, we can define dimensions, metrics, calculated fields, filters, and visualizations to create interactive dashboards. These reports can be shared with stakeholders for monitoring and decision making.

Connectors

What are Connectors?

Connectors are integrated tools that allows looker studio to connect with different data sources such as Google Sheets, Excel files, Databases, Marketing Platforms, and Cloud Services. Looker Studio provides access to more than 200+ data connectors that help bring data from multiple platforms into one reporting environment.

Why do we use Connectors?

Connectors are used to reduce manual data extraction and allows direct communication between the data source and reporting dashboard. They help analysts access updated data and automate reporting processes.

When do we use Connectors?

Connectors are used whenever data required for analysis is stored outside the looker studio. For example, if ecommerce data is stored in Excel file, Google Sheet or a database, a connector is required to bring that data into the looker studio.

How do we use Connectors?

A Connector creates the link between the original data source and Looker Studio. Once connected, Looker Studio can read the available fields, apply calculations, and display the information through dashboards.

Data analysis relevance:

Understanding Connectors is important because analysts frequently work with multiple data sources and need efficient ways to collect and visualize information.

Dimensions and Metrics

Dimensions

Dimensions are qualitative attributes that describe and categorize data. They represent the characteristics or categories used for analysis. In Looker Studio, dimensions are usually represented in green colour.

Metrics

Metrics are quantitative values that measure performance or provide numerical calculations. In Looker Studio, metrics are usually represented in blue colour.

Drill Down Feature

The drill down feature allows users to explore data from a higher-level summary into detailed information by creating a hierarchy between dimensions.

For example: Country > State > City

Why do we use it?

Helps users to analyse data at multiple levels without creating separate reports.

When do we use it?

It's useful when stakeholders want to investigate the reason behind a particular trend or performance change.

Conditional Formatting

Changes the appearance of data based on predefined rules.

Why do we use it?

Helps user quickly identify the patterns, exceptions, and important areas requiring attention.

When do we use it?

Used in dashboards where quick decision making is required.

How does it work?

Rules are created based on conditions. If the condition is satisfied, the formatting is automatically applied.

Data Control Features

Allows dashboard users to interact with reports by changing the data view dynamically

Why do we use it?

To make dashboards interactive and allows users to customize their analysis .

When do we use it?

When multiple users need different views from the same dashboard.

How does it work?

User selections modify the displayed data without changing the original report.

Custom Fields

Allows analysts to create new fields using existing data.

Why do we use it?

To create business specific calculations that are not available directly in the dataset.

When do we use it?

When additional KPIs or calculated values are required.

How does it work?

Custom formulas are created using functions and existing fields.

Concatenate Functions

Combines multiple fields into one.

Why do we use it?

To create meaningful combined information for reporting.

When do we use it?

When data is stored separately but required as one field.

How does it work?

To concatenate function joins text values from different columns.

Running Calculations

Performs cumulative calculations across data points.

Why do we use it?

To analyse cumulative growth and performance over time.

When do we use it?

For trend analysis and time based reporting.

How does it work?

Values are continuously added or calculated across records.

CASE Functions

Allows conditional logic to categorize or modify data.

Why do we use it?

To create categories and business rules.

When do we use it?

When analysis requires classification based on conditions.

How does it work?

It evaluates conditions and returns the required output.

Data Blending

Combines information from multiple data sources into one report.

Why do we use it?

Because business information is often stored across different systems.

When do we use it?

When analysis requires combining datasets that are not available in a single source.

How does it work?

Datasets are connected using a common key.

Community Visualizations

Community Visualizations are custom charts developed by the looker studio community.

Why do we use them?

They provide additional visualization options beyond default charts.

When do we use them?

When standard charts cannot effectively represent the required information.

How does it work?

Custom visual components can be added from the visualization library.

Pivot Tables

Summarizes large dataset by organizing information into rows, columns and values.

Why do we use them?

They help analyze relationships and compares multiple categories.

When do we use them?

For detailed business analysis and summary reporting.

How does it work?

Dimensions are placed as rows/columns and metrics are used as calculated values.

Filters

Filters allows users to restrict the data displayed in a report.

They are used to focus analysis on specific categories, dates, regions, or conditions.

Date Range Control

Date range controls allow users to change the reporting period dynamically. They are useful for analyzing monthly, quarterly, or yearly performance.

Calculated Metrics

Custom measurements created using formulas.

Data Blending Join Keys

Join keys are common fields used to connect multiple datasets.

Report Level vs Chart Level Settings

Report level settings apply changes to the entire dashboard, while chart-level settings apply only to individual visuals. This helps control dashboard customization.

